



Microprocessor & Computer Architecture (μ pCA)

UE24CS251B

Session
- 11



Microprocessor & Computer Architecture (μpCA)

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**Instruction Encoding –
2**

Branch Instruction



Link bit

0 = Branch

1 = Branch with Link

Condition field

Bits	Field	Description
31–28	cond	Condition code (e.g., EQ, NE, AL) – determines if the branch executes.
27–25	101	Identifies branch-type instruction.
24	L	Link bit: 0 = branch only, 1 = branch with link (stores return address in LR).
23–0	imm24	Signed 24-bit immediate offset (word-aligned).

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Branch Target Address= (PC+8)+(offset<<2)

PC=PC+8-the Program Counter (PC) does not hold the address of the currently executing instruction. Instead, due to instruction pipelining, it points ahead by two instructions.

offset is shifted left by 2 (since ARM instructions are 4-byte aligned)

$$\text{Offset} = \frac{\text{Target Address} - (\text{PC} + 8)}{4}$$

Forward Branching



```

.text
00001000:E3A01003  main:  mov r1, #3
00001004:EB000003          bl foo
00001008:E0802001          add r2, r0, r1
0000100C:E0802001          add r2,r0,r1
00001010:E2800001      add r0,r0,#1
00001014:EF000011          swi 0x011
00001018:          foo:
00001018:E3A00002          mov r0, #2
0000101C:E12FFF1E          bx lr
.end

```

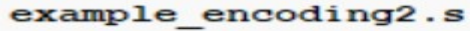
- **Target Address = 0x1018 (address of foo)**
- **PC at Execution = 0x1004 + 8 = 0x100C (because of the ARM pipeline)** Encoded as EB000003.

$$\text{Offset} = \frac{\text{Target Address} - (\text{PC} + 8)}{4}$$

$$\text{Offset} = \frac{0x1018 - 0x100C}{4} = \frac{0xC}{4} = 3$$

[illegible]

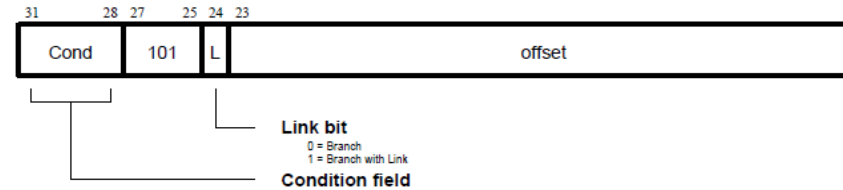
Backward Branching


$$\text{Offset} = \frac{\text{Target Address} - (\text{PC} + 8)}{4}$$

[illegible]

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```
00001000:E3A02004  MOV r2,#4
00001004:E3A06005  MOV r6,#5
00001008:E3A07007  mov r7,#7
0000100C:EA000002  B forward
00001010:E2821004  ADD r1, r2, #4
00001014:E2860002  ADD r0, r6, #2
00001018:E2873004  ADD r3, r7, #4
0000101C:          forward:
0000101C:E2421004  SUB r1, r2, #4
00001020:          backward:
00001020:E2821004  ADD r1, r2, #4
00001024:E2421004  SUB r1, r2, #4
00001028:E0864007  ADD r4, r6, r7
0000102C:EAFFFFFB  B backward
00001030:E0411002  SUB r1,r1,r2
00001034:E0811002  ADD r1,r1,r2
```



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Diagram illustrating the Branch and Link (B/L) instruction format. The instruction is 32 bits long, divided into four main fields:

- Cond (2 bits):** Condition code.
- 101 (3 bits):** Fixed value 101.
- L (1 bit):** Link bit.
 - 0 = Branch
 - 1 = Branch with Link
- offset (26 bits):** Offset value.

$$\text{Offset} = \frac{0x0000101C - 0x00001014}{4}$$
[illegible]

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Diagram illustrating the Branch with Link (BRL) instruction format. The instruction is 32 bits long, divided into four fields:

- Cond** (4 bits, bits 31-28): Condition field.
- 101** (3 bits, bits 27-25): Fixed value 101.
- L** (1 bit, bit 24): Link bit.
- offset** (24 bits, bits 23-0): Offset field.

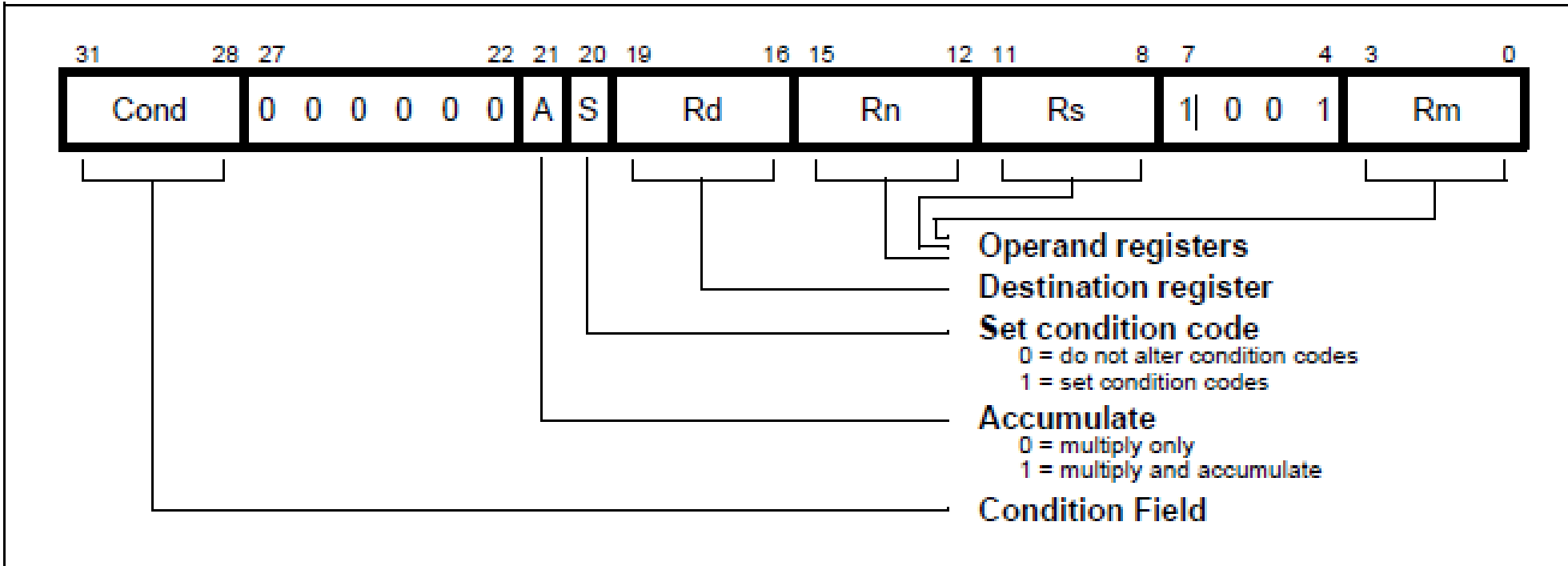
Legend:

- Link bit: 0 = Branch, 1 = Branch with Link

$$\text{Offset} = \frac{0x00001020 - 0x00001034}{4}$$

$$\text{Offset} = \frac{-0x14}{4} = -5$$
[illegible]

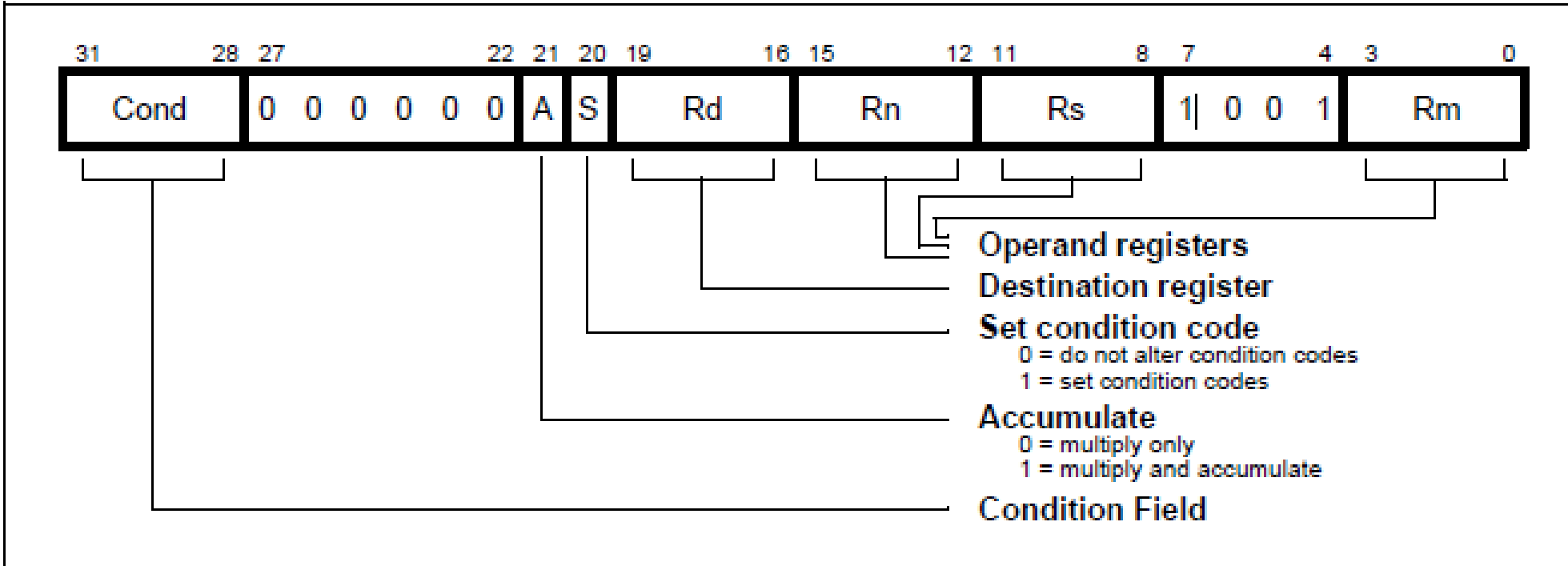
Multiplication Instructions



Ex 1: MUL R0, R1, R2
(MUL Rd, Rm Rs) E0000291

3 1	3 0	2 9	2 8	2 7	2 6	2 5	2 4	2 3	2 2	2 1	2 0	1 9	1 8	1 7	1 6	1 5	1 4	1 3	1 2	1 1	1 0	9	8	7	6	5	4	3	2	1	0	
										A	S	Rd				Rn				Rs								Rm				
1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	0	0	0	1

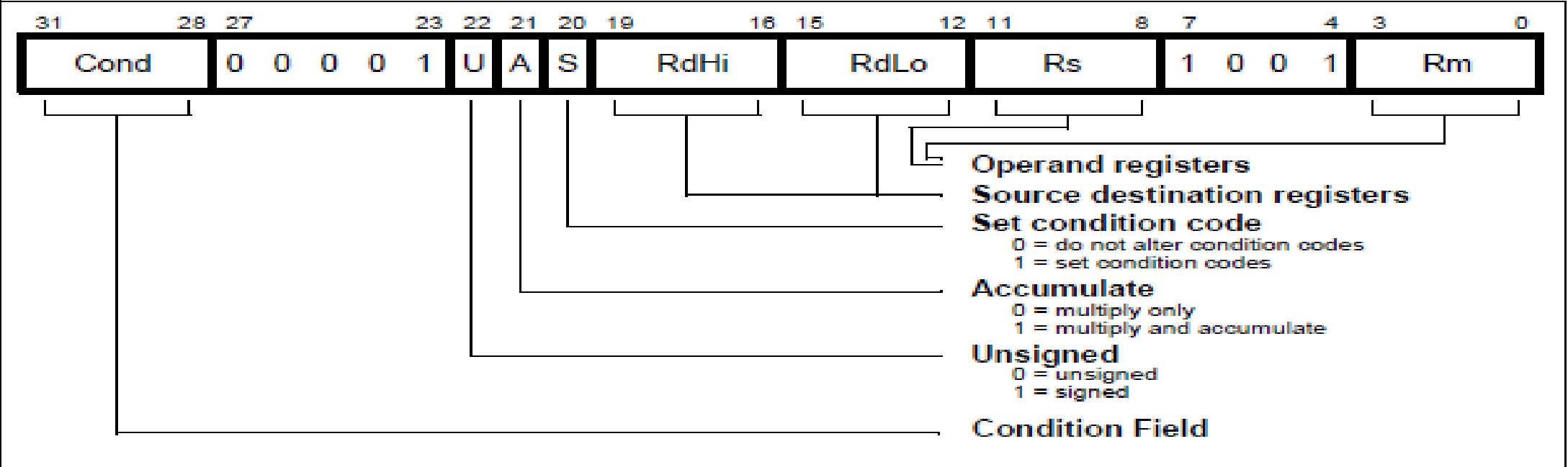
Mul t i pl i cat i on I nst ruct i ons



Ex 2: **MLA** **R0,** **R1,** **R2 ,** **E0203291**
R3 (M A Rd Rm Rs Rn)

3 1	3 0	2 9	2 8	2 7	2 6	2 5	2 4	2 3	2 2	2 1	2 0	1 9	1 8	1 7	1 6	1 5	1 4	1 3	1 2	1 1	1 0	9	8	7	6	5	4	3	2	1	0
Cond										A	S	Rd				Rn				Rs								Rm			
1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	0	0	1	0	0	1

Multiplication Instructions

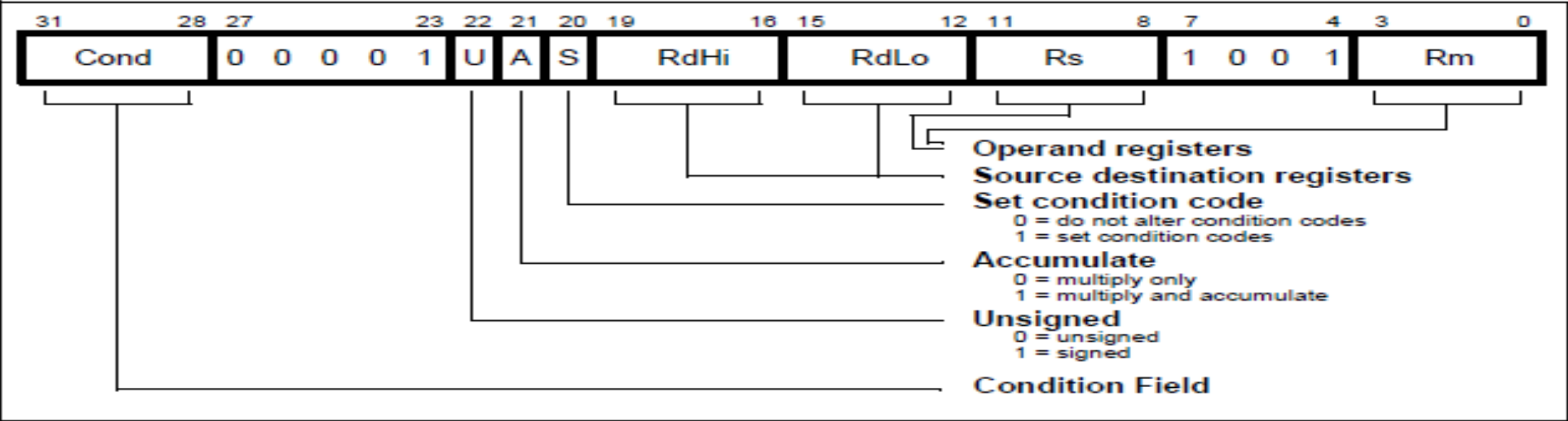


Ex 3: SMULL R0, R1, R2, R3
(SMULL RdLo, RdHi, Rm, Rs)

E0C10392

3 1	3 0	2 9	2 8	2 7	2 6	2 5	2 4	2 3	2 2	2 1	2 0	1 9	1 8	1 7	1 6	1 5	1 4	1 3	1 2	1 1	1 0	9	8	7	6	5	4	3	2	1	0
Cond									U	A	S	RdHi				RdLo				Rs								Rm			
1	1	1	0	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	1	0	0	1	0

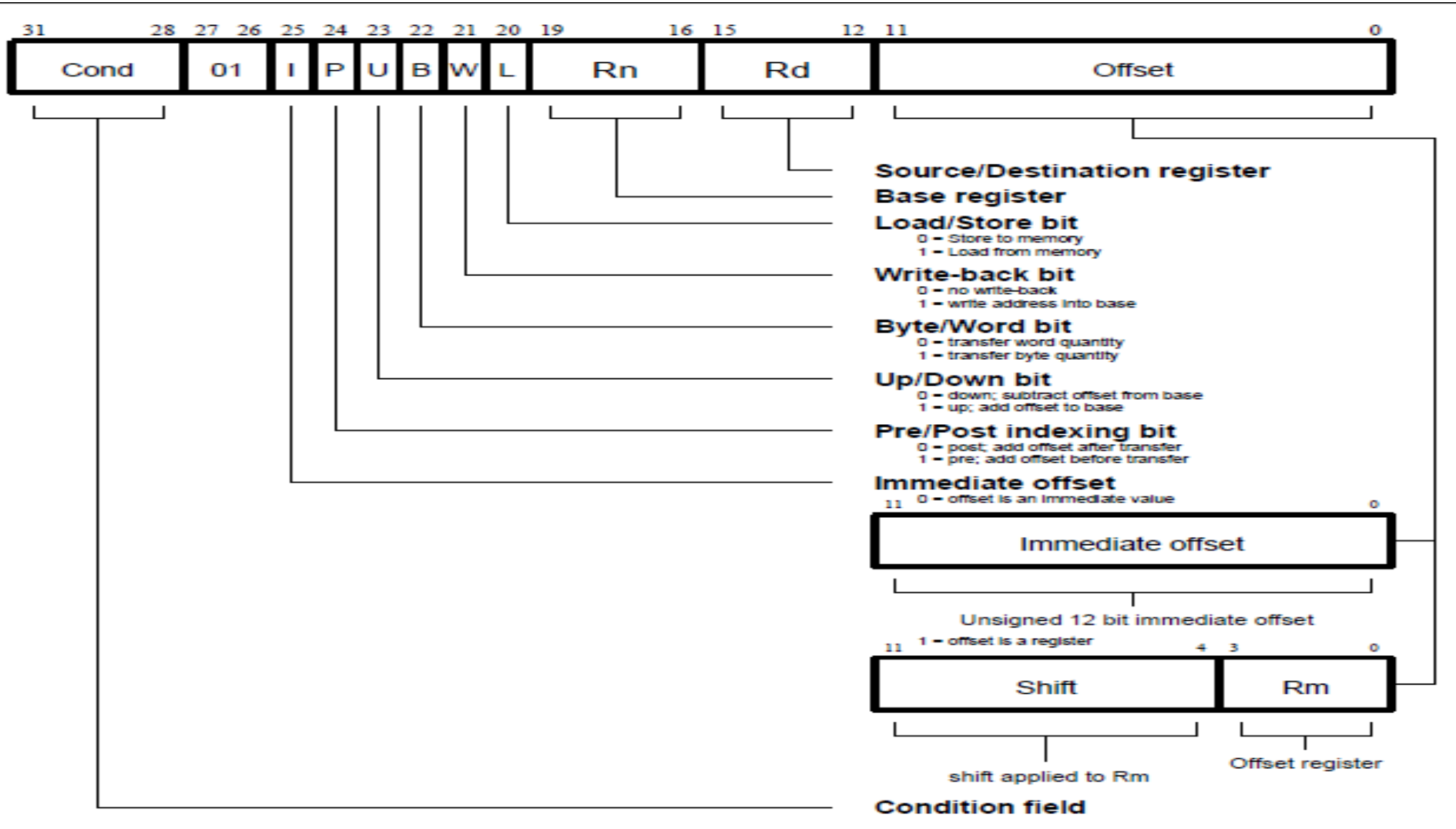
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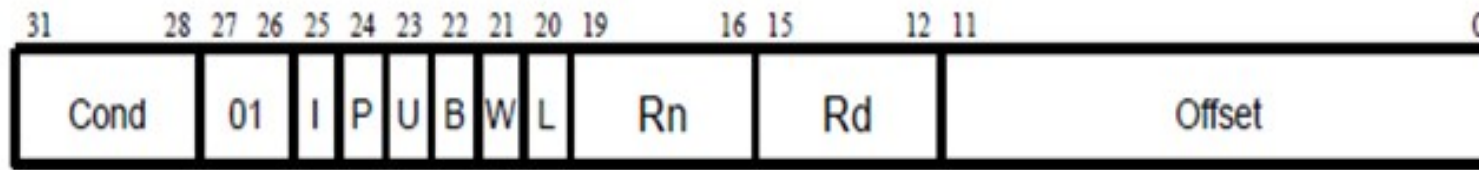
Ex 4: UMLAL R0, R1, R2 , R3 E0A10392

3 1	3 0	2 9	2 8	2 7	2 6	2 5	2 4	2 3	2 2	2 1	2 0	1 9	1 8	1 7	1 6	1 5	1 4	1 3	1 2	1 1	1 0	9	8	7	6	5	4	3	2	1	0
Cond									U	A	S	RdHi				RdLo				Rs								Rm			
1	1	1	0	0	0	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	1	0	0	1	0

Data Transfer Instruction



Data Transfer Instruction: STR



■ Ex:

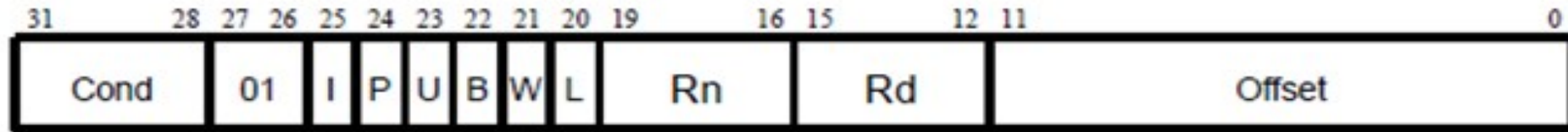
STR R0, [R1]

1110 01 0 1 1 0 0 0 0001 0000 0000 0000
0000

1110 01 I P U B W L 0001 0000 0000 0000
0000

Binary Code: E5810000

Data Transfer Instruction: LDR



■ Ex:

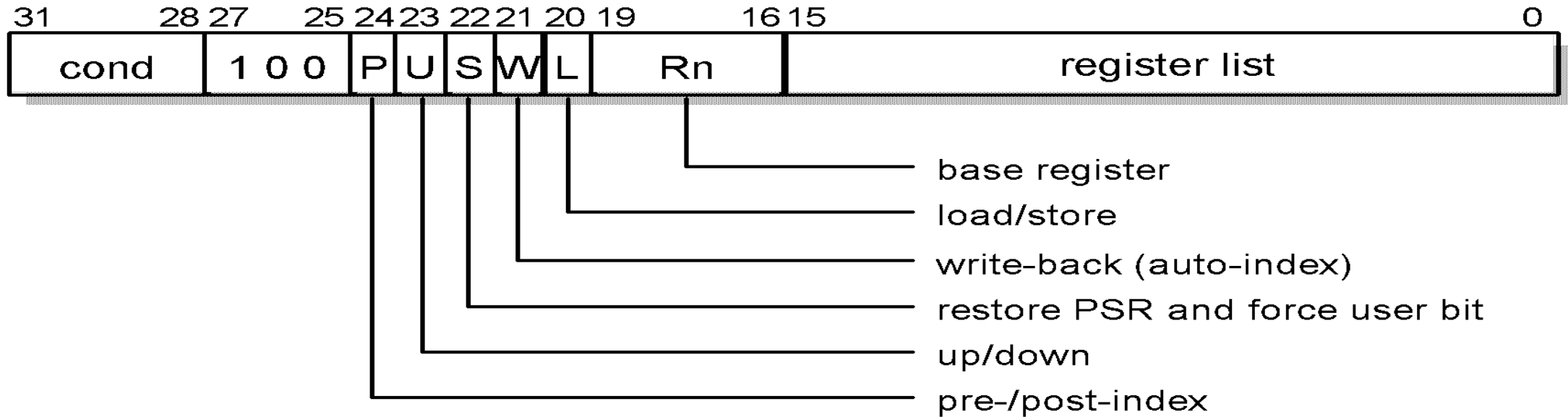
LDR R0, [R1] , R2

1110 01 I P U B W L 0001 0000 0000
 0000 0010

1110 01 1 0 1 0 0 1 0001 0000 0000
 0000 0010

E6910002

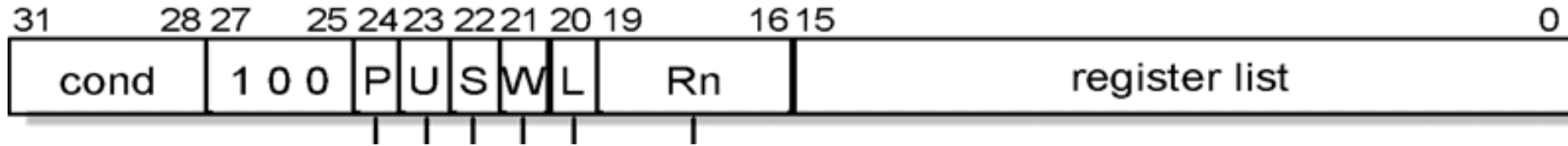
Block Transfer Instructions



EX 1: LDM A R13! , { R0, R5 - R8, R11}

EX 2: STM B R13! , { R8, R4- R6, R12}

Block Transfer Instruction : LDM



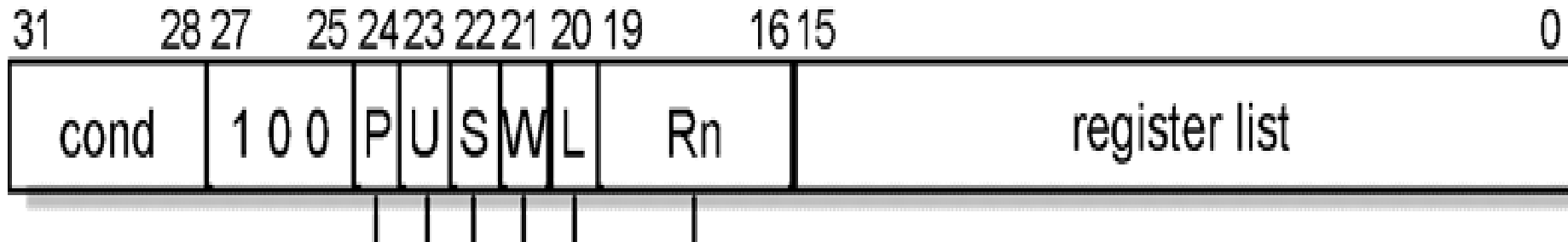
LDM A R13! , { R0, R5 - R8, R11}

1110 100 **P** **U** **S** **W** **L** 1101 0000 **R11** 00**R8R7R6R5**
0000R0

1110 100 **0** **1** **0** **1** **1** 1101 0000 **1** 00**1** **1 1 1**
00001

F8BD09F1

Block Transfer Instruction : STM



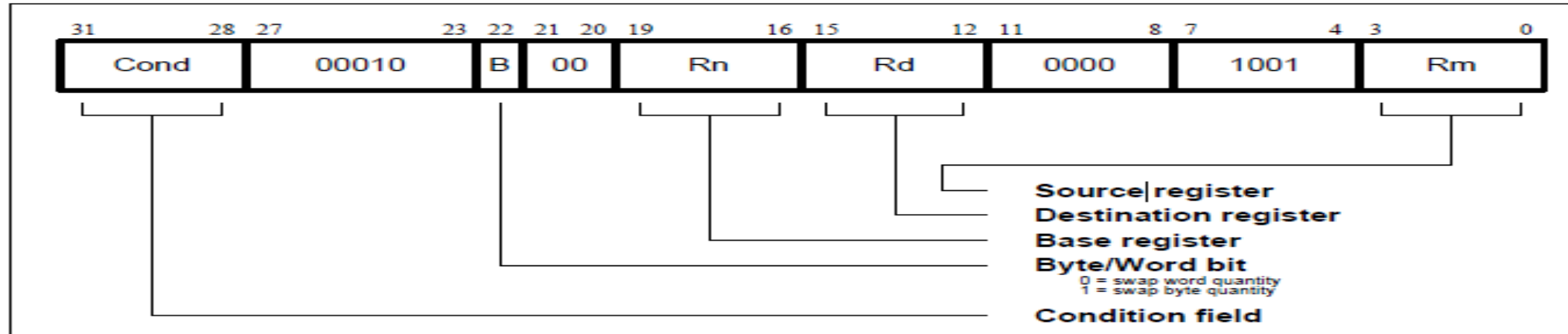
STM B R13! , { R8, R4- R6, R12}

1110 100 1 1 0 1 0 1101 00010
00101110000

1110 100 P U S W L 1101 000R120 00R8
0R6R5R4 0000

E9AD1170

SWAP:



SWP{B}{cond} Rd, Rm, [Rn]

|31-28|27-24|23|22|21|20|19-16|15-12|11-8|7-4 |3-0 |
| cond| 0001 | 0| B | 0| 0| Rn | Rd |0000|1001| Rm |

Eg.: SWP R1, R2, [R3]

1110 0001 0000 0011 0001 0000 1001 0010 E1031092

3	3	2	2	2	2	2	2	2	2	2	2	1	1	1	1	1	1	1	1	1	0	9	8	7	6	5	4	3	2	1	0
1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0										
Cond									B			Rn				Rd												Rm			
1	1	1	0	0	0	0	1	0	0	0	0	0	0	1	1	0	0	0	1	0	0	0	0	1	0	0	1	0	0	1	0

1. In ARM Branch instruction format, what is the function of the Link bit (L) located at bit position 24?
 - A. It determines if the offset is positive or negative.
 - B. It distinguishes between a standard Branch (B) and a Branch with Link (BL).
 - C. It indicates if the instruction allows conditional execution.
 - D. It sets the processor to Thumb mode.

2. Which specific bit in the Multiplication instruction encoding is set to '1' to convert a standard Multiply (MUL) operation into a Multiply-Accumulate (MLA) operation?
 - A. The S bit (Set Condition Code)
 - B. The U bit (Unsigned)
 - C. The A bit (Accumulate)
 - D. The W bit (Write-back)

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Answer – B. It distinguishes between a standard Branch (B) and a Branch with Link (BL).

2. Which specific bit in the Multiplication instruction encoding is set to '1' to convert a standard Multiply (MUL) operation into a Multiply-Accumulate (MLA) operation?

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- D. The W bit (Write-back)

Answer – C. The A bit (Accumulate)

3. In the Single Data Transfer instruction format (used for LDR/STR), which bit indicates whether the data being transferred is a Byte or a Word quantity?
- A. The L bit
 - B. The B bit
 - C. The W bit
 - D. The P bit
4. Which ARM instruction is identified by the specific encoding pattern where bits 27–23 are 00010 and bits 7–4 are 1001?
- A. SWAP (SWP)
 - B. Branch Exchange (BX)
 - C. Store Multiple (STM)
 - D. Multiply Long (SMULL)

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Answer – B. The B bit

4. Which ARM instruction is identified by the specific encoding pattern where bits 27–23 are 00010 and bits 7–4 are 1001?

- A. SWAP (SWP)
- B. Branch Exchange (BX)
- C. Store Multiple (STM)
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Answer – A. SWAP (SWP)



THANK YOU



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Engineering